BERT TEXT SIMILARITY SEARCH ENGINE

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ABSRACT

This project presents the development of a **BERT-based Text Similarity Search Engine** that retrieves contextually relevant documents based on semantic meaning rather than simple keyword matching. Traditional search systems rely heavily on exact word overlap, which often fails to capture the true intent of a query. To address this limitation, this system leverages **sentence embeddings generated using a pre-trained BERT-based model** to represent text in a high-dimensional vector space.

A collection of academic documents from multiple domains such as Physics, Chemistry, Mathematics, Computer Science, and Civil Engineering is used as the dataset. Each document is converted into a numerical embedding using the Sentence Transformers framework. When a user enters a query, it is similarly transformed into an embedding, and **cosine similarity** is applied to measure the similarity between the query and stored document embeddings.

The system then retrieves and ranks the **top five most relevant documents**, providing an efficient and meaningful search experience. This approach demonstrates the effectiveness of modern Natural Language Processing techniques in understanding context and improving information retrieval systems.

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1. INTRODUCTION –

With the rapid growth of digital information, efficient search systems have become essential. Traditional search engines rely on keyword matching, which means they search for exact words present in the query. However, this approach often fails when the wording differs but the meaning is the same.

For example, a query like *"What is artificial intelligence?"* may not match a document containing *"AI is a field of computer science"* due to different wording. This limitation highlights the need for semantic search systems that understand the meaning of text.

This project focuses on building a **semantic search engine using BERT-based embeddings**, which can capture the context and meaning of text. By converting text into numerical vectors and comparing them, the system retrieves results based on similarity in meaning rather than exact words.

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2. OBJECTIVE –

The main objectives of this project are:

- To understand and implement **semantic search techniques**
- To generate sentence **embeddings using a BERT-based model**
- To apply **cosine similarity** for measuring similarity
- To build a simple **command-line interface for searching**
- To retrieve the **top 5 most relevant documents**
- To understand real-world applications of NLP in search systems

3. LITERATURE REVIEW –

- Natural Language Processing has evolved significantly with the introduction of transformer-based models such as BERT, developed by Google. BERT is capable of understanding context by processing text bidirectionally, which improves performance in various NLP tasks.
- Earlier approaches such as TF-IDF and Bag-of-Words relied on word frequency and lacked contextual understanding. These methods often failed in semantic search tasks.
- Recent research shows that **sentence embeddings generated using transformer models** provide better accuracy for similarity tasks. Cosine similarity is widely used to measure similarity between vector representations, making it an effective technique for semantic search systems.

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4. DATASET DESCRIPTION –

The dataset used in this project consists of approximately **50 short academic documents** collected from multiple domains, including:

- **Physics**
- **Chemistry**
- **Mathematics**
- **Computer Science**
- **Civil Engineering**

Each document contains a brief explanation of concepts related to its subject. The data is stored in text or markdown format and organized into files. These documents serve as the knowledge base for the search system.
Although the dataset is relatively small, it is sufficient to demonstrate the working of a semantic search engine.

5. METHODOLOGY –

The system follows a structured pipeline:

Step 1: **Data Loading**

Documents are loaded from a folder and stored as text data.

PURPOSE: Collect all information for processing.

Step 2: **Text Preprocessing**

Text is cleaned by removing special symbols, markdown syntax, and unnecessary spaces.

PURPOSE: Improve data quality

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Step 3: Sentence Tokenization

The text is split into smaller sentences using NLP tools.

PURPOSE: More accurate comparison.

Step 4: Embedding Generation

Each sentence is converted into a numerical vector using a BERT-based model.

PURPOSE: Convert text → mathematical representation.

Step 5: Query Processing

User enters a query, which is also converted into an embedding.

Step 6: Similarity Calculation

Cosine similarity is used to compare query with stored embeddings.

- Higher value = more similar

Step 7: Result Retrieval

Top 5 most similar results are displayed.

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6. RESULTS AND DISCUSSION –

The system successfully retrieves relevant documents based on semantic similarity. Unlike keyword-based systems, it can understand context and provide meaningful results.
For example:

- Query: *"What is AI?"*
- Output: Documents related to machine learning and artificial intelligence

This shows that the system can handle variations in language and still provide accurate results.
However, performance depends on:

- Dataset size
- Quality of text
- Model used

7. CONCLUSION –

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8. FUTURE WORK –

The project can be further improved by:

- Developing a **web interface using Streamlit**
- Using **FAISS for faster similarity search**
- Expanding the dataset for better accuracy
- Supporting additional formats like PDF and DOCX
- Adding multilingual search capabilities

9. REFERENCES –

- Devlin et al., *BERT: Pre-training of Deep Bidirectional Transformers*
- Sentence Transformers Documentation
- Scikit-learn Documentation
- Research papers on semantic search.